Open Research Themes and Problems

UGANDA CHRISTIAN UNIVERSITY
A Centre of Excellence in the Heart of Africa

Detecting and Mitigating Model Drift in Deployed Healthcare Machine Learning Models

Name : Oyet Geoffrey

Access Number: B35603

Registration Number: s25M19/036

Department of Computing & Technology
Faculty of Engineering, Design & Technology

Selected Article and Main Research Question

- Journal Article by:

Rajkomar,A.,Dean,J.,& Kohane,I.(2022)

- **"Machine Learning in Medicine"**

New England Journal of Medicine (NEJM)

Main Research Question:

- How can machine learning be effectively applied in healthcare to improve diagnosis, treatment and patient outcome?

This article reviews how machine learning is transforming healthcare, from disease prediction to clinical decision support. This author highlights impressive technical advances but also acknowledge that "real world healthcare environments are complex and constrained", unlike controlled research settings.

P.O. Box 4, Mukono, Uganda, Plot 67-173, Bishop Tucker Road, Mukono Hill
Tel: +256 (0) 312 350 800 Email: info@ucu.ac.ug Web: https://ucu.ac.ug
Founded by the Province of the Church of Uganda. Chartered by the Government of Uganda

Research Gap

Identified Research Gap

While the article demonstrates the potential of machine learning in healthcare, it does not sufficiently address:

- **How machine learning models can be continuously adapted and validated once deployed in real clinical settings where distribution,worklfows, and patient populations change over time.**

Why This Is a Gap

- Health care data changes continuously(new diseases,protocols,demographics)
- Models often degrade after deployment
- Limited guidance on post-deployment monitoring and updating

Most healthcare ML models performs well during testing but fail silently in practice when conditions change, this deployment gap limits real clinical impact.

P.O. Box 4, Mukono, Uganda, Plot 67-173, Bishop Tucker Road, Mukono Hill
Tel: +256 (0) 312 350 800 Email: info@ucu.ac.ug Web: https://ucu.ac.ug
Founded by the Province of the Church of Uganda. Chartered by the Government of Uganda

Justification: Why the Gap Is Still Relevant

- **Esteva et al. (2019)** demonstrate that although deep learning models show strong performance in controlled settings, significant challenges remain in translating these models into routine clinical practice due to the lack of operational mechanisms for long-term deployment.
- **Kelly et al. (2019)** identify the “last mile problem” in healthcare machine learning, where promising models fail to achieve sustained clinical impact because of inadequate integration, monitoring, and post-deployment evaluation.
- **Topol (2019)** cautions against reliance on static machine learning models in healthcare, emphasizing that evolving patient populations and medical knowledge can lead to performance degradation over time if models are not continuously updated.
- **Nagendran et al. (2020)** provide evidence that deployed medical AI systems can amplify bias and inequities when ongoing performance and fairness assessments are not incorporated into the deployment process.

These studies consistently show that model deployment and sustainability, not model accuracy, remains the core unresolved issue.

P.O. Box 4, Mukono, Uganda, Plot 67-173, Bishop Tucker Road, Mukono Hill
Tel: +256 (0) 312 350 800 Email: info@ucu.ac.ug Web: https://ucu.ac.ug
Founded by the Province of the Church of Uganda. Chartered by the Government of Uganda

Proposed Solution to Fill the Gap

A continuous Learning and Monitoring framework for Healthcare and Machine Learning Models

Core Components

- Automated performance monitoring: Continuously tracks model accuracy and errors in live clinical use.
- Drift detection: Identifies changes in patient data or outcomes that reduce model reliability.
- Periodic retraining: Updates models using recent, validated data when performance decline is detected.
- Human-in-the-loop review: Clinicians review alerts and validate model updates to ensure clinical safety.

Why This Is Novel

- Shifts focus from one-time model training to model lifecycle management
- Aligns ML deployment with clinical realities
- Improves long-term reliability and trust

Instead of assuming models remain valid forever, this approach treats ML systems as living tools that must evolve alongside healthcare systems.

P.O. Box 4, Mukono, Uganda, Plot 67-173, Bishop Tucker Road, Mukono Hill
Tel: +256 (0) 312 350 800 Email: info@ucu.ac.ug Web: https://ucu.ac.ug
Founded by the Province of the Church of Uganda. Chartered by the Government of Uganda

Socialtal,Industry and Domain Impact

Societal Impact

- Safer, reliable healthcare AI: Continuous monitoring prevents outdated or failing models from affecting patient care
- Reduced diagnostic errors: Early detection of performance drift helps maintain accurate clinical predictions.
- Fairer treatment: Ongoing evaluation across patient groups reduces bias and inequitable outcomes.

Industry Impact

- Lower deployment risk: Hospitals can detect and address model failures before they affect operations.
- Regulatory compliance: Supports auditing, transparency, and accountability in AI-driven clinical systems.
- Clinician trust: Human oversight and monitoring increase confidence in AI-assisted decisions.

Domains

- Clinical decision support: Ensures recommendations remain accurate as guidelines and data evolve.
- Disease outbreak monitoring: Adapts models to rapidly changing disease patterns.
- Patient risk stratification: Keeps risk predictions current for timely intervention.

This research helps bridge the gap between promising healthcare AI research and real-world clinical impact, ensuring models remain accurate, fair, and trustworthy over time.

P.O. Box 4, Mukono, Uganda, Plot 67-173, Bishop Tucker Road, Mukono Hill
Tel: +256 (0) 312 350 800 Email: info@ucu.ac.ug Web: https://ucu.ac.ug
Founded by the Province of the Church of Uganda. Chartered by the Government of Uganda

Uganda Christian University

P.O. Box 4 Mukono, Uganda
Tel: 256-312-350800
Email: info@ucu.ac.ug

@UgandaChristianUniversity
@UCUniversity
@ugandachristianuniversity
https://ucu.ac.ug/

A Complete Education for A Complete Person